

OpenBioShield — Assay Design Report

Assay ID: 5019ea22-7d9d-4761-87ed-77592344a9d0 | Generated: 2026-06-10T07:36:46.105765Z | Candidates: 15

Top Primer Candidates

#	Forward (5'→3')	Reverse (5'→3')	Tm F (°C)	Tm R (°C)	GC F (%)	GC R (%)	Prod (bp)	Cov	Thermo	AI	Score
1	ACTCGTCTATCTTCTGCAGGCT	AACCAGGGACAAGGCTCTCC	63.8	64.5	50.0	60.0	118	100.0	62.6	66.6	85.834
2	ACTCGTCTATCTTCTGCAGGCTG	AACCAGGGACAAGGCTCTCC	64.7	64.5	52.2	60.0	118	100.0	54.8	68.4	84.644
3	ACTCGTCTATCTTCTGCAGGC	CCAAAGCCACGTACGAGCAC	62.4	64.8	52.4	60.0	191	100.0	55.1	62.0	83.414
4	TCGTCTATCTTCTGCAGGCTG	CCAAAGCCACGTACGAGCAC	62.4	64.8	52.4	60.0	189	100.0	55.1	61.8	83.384
5	ACTCGTCTATCTTCTGCAGGCT	CCAAAGCCACGTACGAGCAC	63.8	64.8	50.0	60.0	191	100.0	50.8	65.8	83.324
6	CTCGTCTATCTTCTGCAGGCTG	TCCAAAGCCACGTACGAGCA	63.3	65.0	54.5	55.0	191	100.0	52.4	63.6	83.198
7	ACTCGTCTATCTTCTGCAGGCT	TCCAAAGCCACGTACGAGCA	63.8	65.0	50.0	55.0	192	100.0	48.5	65.2	82.744
8	ACTCGTCTATCTTCTGCAGGCT	GCACGTCGCGAACCTGTAAA	63.8	64.4	50.0	55.0	175	100.0	53.6	59.2	82.566
9	ACTCGTCTATCTTCTGCAGGC	TCCAAAGCCACGTACGAGCA	62.4	65.0	52.4	55.0	192	100.0	51.5	61.3	82.556
10	TCGTCTATCTTCTGCAGGCTG	TCCAAAGCCACGTACGAGCA	62.4	65.0	52.4	55.0	190	100.0	51.5	61.2	82.524

Score Weights

Component	Weight
Coverage Score	60%
Thermodynamic	20%
AI Efficiency	20%